NASM VIRTUAL COACHING SPECIALIST (VCS)

Chapter 5: Administering Virtual Assessments

National Academy of Sports Medicine — Evidence-Based Telehealth Curriculum

Chapter 5Administering Virtual Assessments
Lesson 1Virtual Pre-Participation Health Screening

Developing and executing individualized training programs begins with assessing clients. Based on the client's goal, sport, current conditioning, or previous injuries, it is important to develop a baseline with ongoing assessments to achieve outcomes in the safest and most efficient manner possible. This chapter will review how to develop and execute comprehensive assessments in a remote capacity by utilizing technology and other resources available.

Learning Objectives

After completing this chapter, the NASM Virtual Coach will have the ability to do the following:

- **Develop** a virtual assessment toolkit.
- **Identify** safe and effective methods for administering remote assessments.
- **Determine** how to modify remote assessments based on client type or physical readiness.
- **Use** technology to assist in reliable data collection.

Introduction

The role of a fitness professional is to support clients in achieving their health and fitness goals in the safest, most efficient manner possible. Assessments—and subsequently, reassessments—are integral components to this process. Fitness professionals will administer a variety of assessments with clients to establish a starting point for their program and a baseline from which to measure and track progress toward goals over time. With some modifications and additional resources, these assessments can be conducted virtually without compromising safety, utility, and effectiveness.



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While clients are typically able to define their own health and fitness goals, they often lack an objective understanding of their baseline starting point. Considering this, in order to develop an effective, individualized program, fitness professionals must utilize both **objective** and **subjective** assessment tools to gain a better understanding of a client's baseline strengths and areas of opportunity with respect to their goal/s, medical history, exercise history, current levels of fitness and conditioning, recreational or sport activities, and/or previous injuries. Following a phase of analysis, this assessment and reassessment data can be utilized to develop and modify individualized programs, monitor progress toward goals and objectives, and be leveraged to provide reinforcement for adherence and retention.

Administering assessments is common practice for the trained fitness professional. When performed remotely, there are considerations and modifications necessary to maintain safety and **reliability**, which will be discussed in this chapter. Fitness professionals should rely on additional resources and technologies, when applicable, for administering virtual assessments.

Helpful Hint

Navigating a Client's Health and Fitness Journey:

To develop the most efficient route (program), a client must define the intended destination (goal) but also have a precise understanding of the starting location (baseline). As the client is en route to their destination, navigation may recommend a slight detour (program modification made based on reassessment) with the ultimate goal of reaching the destination in the safest and most efficient manner possible.

Objective

An assessment based on facts or direct measurement, such as a quantitative measurement.

Subjective

An assessment based on individual opinion or perception, which lacks the ability to directly measure.

Reliability

The accuracy of a measurement or the degree to which the measurement can be relied upon.

Virtual Assessment Toolkit

Fitness professionals should develop a categorized list of assessments that are feasible, within their credentials and expertise, and can be reliably performed virtually, without physical contact. After developing a comprehensive, categorized assessment list, considerations of application to client type and relevance, access to specific equipment and/or technology, and modifications based on previous injuries or physical limitations will help organize and standardize the assessment process.

While not exhaustive, **Table: Virtual Assessments** lists common assessments that can be administered virtually. A fitness professional should assess their own skills, expertise, resources, and feasibility/practicality when developing an assessment list for their own coaching business.



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Virtual Assessments

Assessment Type	Description	Examples
Questionnaire/Survey	A specific set of questions to assess an individual's goals, health status, or lifestyle habits	PAR-Q+ Health History Questionnaire Dietary Assessment Forms
Physiological	An evaluation of certain physical and physiological characteristics	Heart rate Blood pressure Cardiorespiratory Fitness Anthropometric
Posture, Movement, and Mobility	An evaluation of an individual's movement pattern quality, neuromuscular control, and/or range of motion	Static Posture Apley scratch test Overhead squat Single-leg squat Pushing/pulling

Muscular Endurance	An evaluation of muscular strength with respect to time or repetitions until fatigue or compensation	Push-up Plank test Wall sit test

Fitness professionals should then develop a general schedule of assessment frequency (**Table: General Schedule of Assessment Frequency**), which may vary based on the client type or perceived progress. Reassessment provides the opportunity to show objective and quantifiable progress to maintain client adherence and motivation, but reassessing too frequently can also lead to discouragement if progress is not evident. For example, a weight-loss client may reassess scale weight on a weekly or biweekly basis but perhaps only take circumference measurements once per month.

General Schedule of Assessment Frequency

Assessment Type	Description	Resources (Required and Recommended)	Frequency of Reassessment
Scale weight	Assessment of physical body weight, measured in lbs	Digital or manual (analog) scale	Determined on client-by-client basis
Resting heart rate	Assessment of heart rate at rest, measured in beats per minute (bpm)	Instructions Clock or stopwatch	Bi-weekly Continuous if wearable device available
Overhead squat	Functional assessment of kinetic chain neuromuscular control and compensatory mechanisms	Instructions Video: live stream and/or recorded	Monthly As needed to assess progress and modify program
Push-up test	Assessment of upper body strength and endurance	Instructions Video: live stream and/or recorded	Monthly As needed to assess progress and modify program

General Schedule of Assessment Frequency

Technology and Resources

According to the Pew Research Center (2019), approximately 96% of Americans own a cell phone and nearly three-quarters of U.S. adults own a personal computer. These devices are often equipped with valuable technology for facilitating remote assessments, including high-resolution cameras, video conferencing applications, a timer, stopwatch, and email access.

While a fitness professional should not rely on a client having access to such technology, the ability to use these resources when available can significantly expedite and expand the range of assessments conducted while improving interpretation of the data. See **Table: Types of Technology** for types of technology the fitness professional might use.

Types of Technology

Technology	Technology Description
 	Conducting movement assessments via a video conferencing session may allow for better communication of instructions, improve client comfort, and be recorded for interpretation or side-by-side comparison.





Using email or text messaging for daily communication, or in addition to a twice-a-week training session, may help build and maintain a trusting client–trainer relationship while improving a fitness professional's ability to manage more clients.



Android and Apple cell phones often come pre-equipped with health applications (Samsung Health or Apple Health) with features that allow clients to measure, track, and even share their data, including workouts, caloric intake, weight, and heart rate. Cell phones can also be used to facilitate more personal communication with clients.



Cell phones, computers, or standalone cameras provide the ability to record physiological assessments for later interpretation so fitness professionals can dedicate real-time efforts toward motivating, coaching, and facilitating a safe and reliable virtual assessment.

In addition to the connected devices, there are many software platforms available for fitness professionals. These include but are not limited to survey, nutrition, and all-in-one training applications that can aggregate, analyze, and trend assessment data over time while improving the efficiency and individualization of program development. A spreadsheet is an alternative option in addition to software platforms to log client data and stay organized.

Table: Example Applications and Platforms provides a few examples of highly rated applications and platforms used by fitness professionals but it should not be considered a comprehensive or exhaustive list. Fitness professionals should do their own research to determine the best platform solution with regard to client type, usability, and cost.

Example Applications and Platforms

Survey Tools	Nutrition Tracking	All-in-One
Typeform	MyFitnessPal	NASM Edge
Google Forms	MyNetDiary	TrueCoach
SurveyMonkey	MyPlate Calorie Counter	My PT Hub

Administering Pre-Participation Health Screening

Important components of program development include safety and sustainability. In order to develop individualized programs that address these components, fitness professionals should always administer some type of pre-participation health screening as a portion of the intake process. There are various types of health screening tools available, which can be efficiently conducted via email, phone call, or video conferencing. Fitness professionals may standardize the pre-participation health screening process or determine which tools to use on a client-by-client basis.

To begin the intake process, the fitness professional should do the following:

1. Assess his/her client’s experience, comfort, and access to technology, or ask directly which method they would prefer.
2. Administer the pre-participation health screening. In general, some form of pre-participation health screening is recommended for all fitness professionals, but it may not be necessary to perform all of the listed pre-health screening tools.
3. Organize the pre-participation health screening information for future access with consideration to client privacy. Instead of saving client files on a desktop, fitness professionals may consider a cloud storage account. Not only do these platforms require a password and additional security, but they also provide accessibility from any connected device.

In addition, there are many online tools and websites that allow professionals to build custom surveys, automatically share by e-mail at a predefined frequency, and produce summary reports.

Coach's Corner

While saving client information online, be sure to keep this information secure and consistent with HIPAA (1996) policies.

Physical Activity Readiness Questionnaire for Everyone (PAR-Q+)

The Physical Activity Readiness Questionnaire for Everyone (PAR-Q+) is a standardized assessment of a client's readiness to engage in physical activity. Fitness professionals should avoid making assumptions or relying on aesthetic characteristics alone when determining physical readiness for any type of program or intervention. Due to the personal and sensitive nature of the questions, the PAR-Q+ is generally self-administered with minimal guidance. Instructions are included in the PAR-Q+, but a fitness professional should consider providing a brief overview and instructions.

The PAR-Q+ consists entirely of yes or no answers, with instructions to complete additional sections based on responses. In order to be considered valid, the PAR-Q+ does require a participant signature. For this reason, clients may consider printing and scanning the completed document, which may be done with a scanner or personal camera and sent via email or text message (**Figure: Technology for PAR-Q+**).



FigureTechnology for PAR-Q+
Food for Thought

PAR-Q Versus PAR-Q+

Among the most accepted and widely used health screening surveys is the PAR-Q+. The original PAR-Q, created by the British Columbia Ministry of Health and the Multidisciplinary Board on Exercise, was developed as a standardized assessment to gauge an individual's readiness to perform physical activity. Based on feedback from individuals engaging in physical activity, fitness professionals, and medical physicians, researchers found that the original PAR-Q had limitations in terms of utility and effectiveness across the broad population, citing that the established health benefits of participation far outweigh the risks (Bredin, Gledhill, Jamnik & Warburton, 2013).

Whereas the PAR-Q unnecessarily restricted exercise for populations who would benefit from physical activity, the revised PAR-Q+ was developed to be more inclusive, with the + indicating "For Everyone."

2020 PAR-Q+ Template

Lifestyle and Health History Questionnaire

The Lifestyle and Health History Questionnaire is intended to assess a client's baseline habits, goals, and readiness to safely engage in physical activity. Fitness professionals should provide a brief overview and instructions, if self-administered, or consider guiding the client through each category and section via video conference or phone (**Figure: Technology for Questionnaire**).



FigureTechnology for Questionnaire

The Lifestyle and Health History Questionnaire is separated into six distinct categories:

1. Exercise
2. Diet
3. Lifestyle
4. Occupation
5. Recreation
6. Medical

It may not be necessary to use this entire questionnaire if the PAR-Q+ has been completed; however, fitness professionals should use some alternative methods to obtain user-defined goals and lifestyle habits that may impact program outcomes. Additionally, fitness professionals should consider asking further questions based on the client's lifestyle, occupation, or goals. For example, if the client rates a particularly high level of work-related stress, questions related to an average work week may provide deeper transparency into how to develop an effective program with realistic expectations.

Lifestyle and Health History Questionnaire

Dietary Assessment Forms

Dietary assessment forms are utilized to identify and monitor nutritional patterns, including consumption of all food, beverages, and supplementation. The purpose of nutritional assessments is not to discourage a client but, rather, help bring attention to nutritional habits that might discourage progress. Fitness professionals may decide to use all, some, or none of these dietary assessment forms based on their qualifications and expertise.

24-Hour Recall

The 24-hour recall is intended to rapidly assess a client's perception and memory of nutritional intake of the previous 24 hours, including all food, beverages, and supplements. The fitness professional should prompt client reporting of the following:

- Food Item Description
  - Type of food or beverage
  - Method of preparation (home cooked, leftovers, pre-made, restaurant, etc.)
  - Brand name or restaurant
- Serving Size/Amount/Portion Description
- Time Consumed
- Location Consumed and Occasion
  - i.e., kitchen table, living room, restaurant, etc.
  - Why? i.e., hungry, bored, post-exercise fueling, etc.



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Normally performed as an in-person interview, it is recommended that fitness professionals conduct the 24-hour recall via phone call or video conference (**Figure: Technology for 24-Hour Recall**). Because this assessment should be based on the client's memory, communications methods such as text messaging or email should be avoided.



FigureTechnology for 24-Hour Recall

Critical!

Understanding Limitations of 24-Hour Recall:

Self-administered nutrition assessments, including the 24-hour recall, may be subject to underreporting of energy intake, particularly in certain subgroups of the population including those with a high BMI (Johansson et al., 2001). In general, fitness professionals should use this assessment tool to improve client self-awareness and to improve nutritional habits through comparison of 24-hour recall logs over time.

24-Hour Recall Worksheet

Usual Intake Worksheet

The usual intake worksheet is intended to gain a broad overview of nutritional habits in general, not limited to just the previous 24-hours. As opposed to unstructured recall, the fitness professional should ask specific questions related to the consumption frequency of various food and beverage categories. A common assessment method includes structured question and response choices as follows:



FigureTechnology for Usual Intake Worksheet

- Over the previous year, how often did you consume alcohol?
    - Never
    - 1 time per month or less
    - 2–3 times per month
    - 1–2 times per week
    - 3–4 times per week
    - 5–6 times per week
    - Daily
    - Multiple times per day

Important intake categories to assess include but may not be limited to water, caffeine, foods and beverages containing a significant amount of sugar, and specific questions to gauge intake of specific micro- and macronutrients. Another method is to provide a list of time segments throughout the day (early morning, morning, midday, etc.) and have a client describe their normal intake in their own words.

The usual intake assessments can be performed via phone call, video conference, or email (**Figure: Technology for Usual Intake Worksheet**).

Usual Intake Worksheet

Interpreting Results

Fitness professionals should thoroughly analyze clients' nutritional habits to gauge caloric intake and potential macro- and micronutrient deficiencies. Manually calculating all of these foods can be a tedious task; therefore, it is recommended that fitness professionals use a nutritional tracking software to analyze and interpret the nutritional assessment forms. The NASM Edge app (**Figure: NASM Edge app**) also offers a macronutrient calculator. The purpose is to understand individual caloric requirements and nutritional habits that may limit program efficacy and to develop meaningful and sustainable interventions for long-term success.



FigureNASM Edge app

Administering Physiological Assessments Virtually

In addition to survey information, an effective program should be developed based on individualized physiological data. Physiological measurements are those that objectively quantify certain characteristics, including body weight, circumference measurements, and vital signs. Fitness professionals should focus on education and committable habits to help clients understand the importance of accurately measuring physiological data. Leveraging cost-effective and readily available technology will improve data reliability and avoid human error in data collection and/or reporting.

Physiological Assessments

It is important to understand that physiological assessments are sensitive to the external environment, influenced by physical exertion, and may fluctuate significantly throughout the day. Fitness professionals should consciously avoid confounds during physiological assessments. As a general rule, physiological assessments should be performed in the following ways:

- First thing in the morning, ideally at the same time each day, in the same outfit
- In the same position each time – Results can be significantly different in a seated versus standing position.
- No earlier than 2 hours following exercise or a stressful event
- No earlier than 1 hour after consuming caffeine
- Using the same data collection method, instructions, and equipment

Fitness professionals should leverage technology and resources to expedite and improve the reliability of virtual physiological assessments. In addition to comprehensive—yet simple and easy-to-understand—written instructions provided via e-mail or text message, fitness professionals may consider pre-recording instructional videos or dedicating time to walk clients through these assessments with live video conferencing, coaching, and demonstration (**Figure: Technology for Physiological Assessments**).



FigureTechnology for Physiological Assessments

Once prepared, these instructional videos can be shared via e-mail or a preferred communication strategy to improve the efficiency of conducting virtual physiological assessments.

Fitness professionals should avoid generic e-mails and pre-recorded videos by taking the time to customize the e-mail and video introductions, directly referencing the client's name, program goal, and/or modifications that were made based on injury history or health conditions (**Figure: Customized Video Introduction**).



FigureCustomized Video Introduction

To facilitate the communication and monitoring of assessment results, fitness professionals should create a standardized process based on the resources available. For example, providing a template in Google Sheets or another spreadsheet program offers a secure and efficient method compared to manual reporting of results by text or e-mail. Based on the assessment frequency, fitness professionals can organize these sheets and schedule automated reminders to keep clients accountable and prompt and monitor compliance. When applications such as Samsung Health or Apple Health are available, clients have the ability to manually enter results, including heart rate, blood pressure, measurements, and scale weight and, with a single click, they can share their results with friends, family members, and/or health and fitness professionals.

Resting Heart Rate

Resting heart rate, measured in beats per minute, is a common physiological assessment to non-invasively gauge an individual's health and fitness. According to a study of nearly 3,000 male subjects followed over 16 years, researchers concluded that elevated resting heart rate is a risk factor for mortality independent of physical fitness and other major cardiovascular risk factors (Jensen et al. 2013). For this reason, fitness professionals should consistently assess resting heart rate to monitor progress or changes in health status.

Fitness professionals should take the time to guide their clients through a baseline resting heart rate assessment through visual demonstration when possible, which can be accomplished through pre-recorded instructional videos sent by e-mail or a live video conference (**Figure: Technology for Resting Heart Rate**).



FigureTechnology for Resting Heart Rate

To measure RHR via manual palpation (**Figure: Measuring Palpation**), consider the following:

- Have the individual sit or lie comfortably for several minutes prior to testing.
- The RHR is measured via fingertip palpation – Use the index and middle fingertips rather than the thumb, as it has its own pulse.
- The fingertips are placed directly over the artery and with lightly applied pressure.
- Once a pulse is identified, count the pulses for 60 seconds.
- Note that there is no universal consensus on whether the first beat measured should commence with ‘0’ rather than the number ‘1' because cardiac cycles are being counted. Whichever is decided on, do it consistently.
- When measuring exercise heart rate, this count can be measured over a 10-second interval and then multiplied by six.



FigureMeasuring Palpation



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Another method for regularly assessing a client’s resting heart rate, when applicable, is through a wearable device or fitness tracker. These devices often collect heart rate data while the client is sleeping to accurately assess and trend an individual’s resting heart rate over time. Additionally, many cell phones have the ability to reliably measure heart rate using the camera light. Access to these heart rate measuring tools can often be determined by visiting the pre-installed health app (Apple Health or Samsung Health) on most newer devices.

Heart Rate Recovery

Heart Rate Recovery (HRR) is a measurement of a client's heart rate after a defined period of time following intense exercise. For example, if a client were to perform an exercise until they achieve an active heart rate of 160 beats per minute, and then, after 1 minute of no exercise, their heart rate returns to 145 beats per minute, this client would have a heart rate recovery of 15 beats per minute. In general, a higher value for heart rate recovery is preferred, as this is correlated with improved fitness and overall cardiovascular health (Dimkpa, 2009).

The method for assessing heart rate recovery should be determined based on the client's baseline physical fitness and activity level and should always be performed under the guidance of a trained fitness professional. For example, a less-active or older client may be guided through an incremental walk test, where the intensity or speed is gradually increased and the rate of perceived exertion is consistently assessed by the fitness professional. To ensure safety, it is recommended that the fitness professional perform this assessment via video conference or phone call to guide the user through the protocol and assist with recording the heart rate measurements (**Figure: Technology for Heart Rate Recovery**).

If the client is a well-trained athlete, a fitness professional may consider a more rigorous protocol, such as a functional threshold power assessment. In this population, the client may be more likely to have a heart rate monitor and can accurately and safely perform this assessment without much guidance, although it is still recommended that this be conducted via video conference.

In both cases, the method for assessing heart rate recovery should be standardized. Following exercise, the client may stand up or sit down during the recovery period, but the fitness professional should document this and ensure that future reassessments are performed under the same conditions as this may have an effect on the results.



FigureTechnology for Heart Rate Recovery

Blood Pressure

Based on the results of the PAR-Q+ and Health History Questionnaire, it may be necessary to prompt clients to regularly assess their blood pressure. There are many affordable over-the-counter blood pressure devices available, including cuff-based devices worn at the wrist or upper arm. While wrist devices have advantages in terms of size, cost, and fitting, it is important to consider data integrity of such an important physiological assessment. In a population-based study, researchers found that "use of wrist devices for home self-measurement leads to frequent detection of falsely elevated blood pressure values" (Casiglia et al.2016). For this reason, fitness professionals are encouraged to recommend traditional upper-arm monitors.

Prior to assessing blood pressure, clients should be instructed to sit at rest for a minimum of 5 minutes. The following steps are important to reliably measure blood pressure, regardless of the device manufacturer:



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FIGURETechnology for Blood Pressure

1. Remove clothing or anything preventing direct contact between the cuff and skin.
2. Apply the cuff, making sure that the indicator line is placed directly over the artery.
3. Ensure that the elbow is placed comfortably on a stationary object (such as a desk) with the forearm horizontal across the body. The wrist bone should be at heart level.
4. Avoid excessive movement during the blood pressure measurement.

Fitness professionals should take the time to guide their clients through baseline blood pressure monitoring via video conferencing or pre-recorded instructional videos to ensure instructions are closely followed to optimize data reliability (**Figure: Technology for Blood Pressure**). If abnormal values are reported, fitness professionals should consider the live video conference method with step-by-step instructions and feedback to ensure reliable values are obtained.

Anthropometric Assessments

Anthropometric assessments are quantitative measurements of the physical body with commonly assessed components, including height, weight, body mass index (BMI), and body circumference (Casadei & Kiel, 2020). These measurements are generally simple to obtain and can be used as a tool to monitor progress, provide reinforcement and motivation, and dynamically adjust programs.

Scale Weight

Scale weight is among the most common assessment tools available. While it may be simple to instruct a client to step on a scale and write down a number, there are many considerations that impact the reliability of scale weight measurements. The most important consideration for scale weight is consistency.

Due to daily fluctuations in weight throughout the day based on drinking fluids, consuming food, and using the restroom, that simple number on the scale can quickly turn into a less-meaningful tool to assess progress and may lead to discouragement. Fitness professionals may consider averaging client weight over 2 to 4 weeks to account for these normal fluctuations and for a more reliable measurement of change in scale weight over time, particularly in women who may experience weight gain during the menstrual cycle. To facilitate the communication and monitoring of scale weight results, fitness professionals may create a Google Sheet or use another spreadsheet program for immediate manual entry following measurement. Additionally, fitness professionals can instruct clients to immediately report results via e-mail or text message (**Figure: Technology for Scale Weight**).



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FIGURETechnology for Scale Weight

In general, a fitness professional should instruct their client to weigh themselves first thing in the morning, prior to the first meal, wearing minimal clothing or the same outfit. Fitness professionals should consider placing less emphasis on scale weight and shifting validation of program success to other assessments that accurately reflect progress, such as photos and circumference measurements.

Critical!

Placing too much emphasis on scale weight can be counterproductive. According to Dionne & Yeudall (2005),

Attention focused on one's current, and presumably undesirable, body size can be motivating to those who want to modify a pattern of unhealthy behavior. For others, this feedback may result in psychological distress and could lead to outcomes that are, in fact, counter to a weight loss strategy, such as attrition from programs, misreporting of dietary intake, and emotional states associated with binge eating.

A fitness professional should consider the client type and program goal to determine how often measurements such as scale weight should be recorded as well as the value to the functional goal of the client in general.

Body Mass Index

Body mass index (BMI) is a standardized classification tool for indirectly assessing body fat based on a client's height and weight. BMI classification categories include underweight, healthy weight, overweight, and obese with correlations related to an individual's risk of heart disease, high blood pressure, Type II diabetes, and other health conditions (National Institute of Health). For this calculation, it is important to input accurate values for height and weight.

To measure height, a client should be instructed to do the following:

1. Stand flat with heels, back, buttocks, and shoulders against a wall.
2. Keep the head straight and chin level to the ground.
3. Mark a line on the wall directly at the top of the head, not including the hair.
4. Step away from the wall and then use a tape measure to precisely measure the distance from the ground to the drawn line.



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BMI can be determined using several methods, including mathematical formulas available in both metric and imperial units and various online calculators. There are two formulas to calculate BMI. It is calculated by either dividing a person's weight in kilograms by the square of their height in meters or dividing their body weight in pounds by the square of their height in inches and multiplying by 703.

- Metric Formula: BMI = weight (kg) ÷ [height (m)]^2
- Imperial Formula: BMI = 703 x weight (lbs.) ÷ [height (in)]^2

To avoid errors in calculation fitness professionals should find a reliable BMI calculator app or online tool, such as the Adult BMI Calculator (Centers for Disease Control and Prevention, 2020), which automatically classifies the client's BMI. If the client does not have a scale, purchasing one that reports BMI may be encouraged. Similarly, these tools require accurate inputs of height and weight.

To facilitate the communication and monitoring of BMI results, many smart scales offer the ability to manually share the results and trend graphs with a single click. If this resource is not available, fitness professionals should add a column for BMI input into a prepared Google Sheet or other spreadsheet program or provide instructions to e-mail and/or text results immediately following measurement.

Coach's Corner

It is important to note that BMI has limitations in certain populations, including individuals and athletes with higher bone density and muscle mass.

Waist-to-Hip Ratio

Measuring waist-to-hip ratio (WHR) is another cost-effective and simple way to perform an assessment that provides insight into a client's body composition and the ability to show progress over time through scheduled reassessment. In a study published by the *Annals of Internal Medicine*, researchers found that obesity defined by WHR was associated with higher mortality rate than body mass index (Sahakyan et al., 2015).

The only equipment required to assess waist and hip circumference is a measuring tape, which is typically found at local general merchandise, pharmacy, or online retailers. A measuring tape should not be confused with a tape measure, which is more rigid and will not snugly wrap around the body.

To measure waist and hip circumference, it is important to provide guidance or instructional images to the client to ensure these areas are accurately and precisely measured. For initial baseline measurements, anthropometric assessments should be conducted via video conferencing to ensure the appropriate physical landmarks are adhered to. Follow-up assessments may be conducted via phone call or self-administered, but fitness professionals should assess for abnormal values and revert to video conferencing as needed.

To accurately measure waist circumference, do the following:

1. Identify the narrowest point of the torso below the rib cage and above the iliac crest (top portion of the hip bone) at the hips.
2. Take this measurement as close to the skin as possible.
3. Be sure the tape runs horizontally around the entire body circumference.
4. Keep the tape flat and avoid any twisting.
5. The tape should be pulled snugly but not to the point of causing an indentation in the skin.
6. Take the waist measurement at end-tidal volume, following normal expiration.
7. Record scores to nearest millimeter or 1/16″.



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To accurately measure hip circumference, do the following:

1. Stand straight with feet together and locate the widest portion of the hips, often at the level of the mid-buttocks.
2. Wrap the measuring tape snugly around this area, ensuring that the measuring tape is level and not angled. The measuring tape should not be pulled so tight as to compress the skin.
3. Record the hip measurement.
4. Take the measurement at least twice to ensure accuracy. If the values do not match, perform until repeatable measurements are obtained.

To calculate waist-to-hip ratio, simply convert the measurements to centimeters (cm) and divide the waist measurement by the hip measurement. The World Health Organization (WHO, 2008) defines abdominal obesity as a WHR above 0.90 in males and above 0.85 in females (2008) **(Figure: Hip Circumference)**.



FigureHip Circumference

To personalize and optimize the efficiency of these assessments, consider filming a short instructional video of how to properly take waist and hip circumference measurements. These pre-recorded instructional videos can be shared with virtual training clients via e-mail or text message to expedite the assessment process. To facilitate the communication and monitoring of results, fitness professionals may create an additional column in Google Sheets or another spreadsheet program for manual input or prompt clients to immediately send results via e-mail or text message following measurement.

Circumference Measurements

Depending on the client type and program goal, a fitness professional may consider a more comprehensive assessment of circumference measurements. Once again, it is important to provide guidance and instructional images or videos to ensure the measurements are accurately obtained at the correct physical location. Circumference measurements should not be utilized to assess fatness but, rather, to identify progress or changes in overall girth. The placement of the measuring tape for each circumference measurement is as follows:

Circumference Measurement Sites

Site	Description of Location	Placement of Measuring Tape
Neck	Measure just below the larynx (across the Adam's apple for men).	
Chest	Measure across the fullest part of chest.	
Waist	Measure at the narrowest point of the waist, below the rib cage and just above the top of the hipbones. If there is no apparent narrowing of the waist, measure at the navel.	

Hips	With feet together, measure circumference at the widest portion of the buttocks.	
Thigh	While standing, legs 10 cm (4″) apart, measure the largest circumference immediately below the gluteal fold.	
Calves	Measure the calf at its fullest part, taken between the ankle and the knee.	
Arm	Measure the upper arm at its fullest part, taken at midpoint between the shoulder and elbow.	

In general, all circumference measurements should be recorded at least twice to ensure accuracy. If the two measurements are not identical, continue to assess until repeatable measurements are obtained. For circumference measurements including the thighs, calves and biceps, a fitness professional should determine if measurements will be taken on both sides or limited to one side only. If circumference measurements are taken on only one side, it is important to document this to ensure consistency when performing reassessments.

As with the waist and hip measurements, it is recommended that the fitness professional takes the time to create personalized demonstration videos for the client on how to properly record circumference measurements. Clients can document and share these results in a standardized template provided by the fitness professional or immediately share results by e-mail or text message following measurements.

Bioelectrical Impedance Analysis (BIA)

Bioelectrical Impedance Analysis (BIA) is a reliable measurement tool for assessing additional anthropometric measurements, including fat mass, skeletal muscle mass, total body water, and protein and mineral content. Obtaining these additional measurements will allow fitness professionals to go beyond simple scale weight monitoring to objectively assess changes in body fat and muscle to show meaningful progress and optimize client adherence and retention.

BIA works by passing a small and precise frequency of electricity through the body and measuring the impedance or resistance caused by specific types of body composition. For example, this electricity will flow smoothly through water, but will have higher levels of resistance when passing through fat tissue. This assessment requires physical contact on electrode sites, most commonly both feet (two contact points) when using a scale incorporating BIA technology, but often includes the hands (four contact points) in more advanced devices as well.



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FIGURETechnology for BIA

Depending on the program goal, a fitness professional may consider recommending that their client visit a location that offers BIA testing or purchase an at home BIA scale for regular assessment of body composition. In both instances, the results of single or multiple assessments may be shared as a PDF via e-mail or directly through a reporting software (**Figure: Technology for BIA**). While BIA is considered a generally safe assessment, it is important to note that there are contraindications, including pacemakers, metal implants, and pregnancy. In addition, BIA assessment should be done under similar circumstances, if possible, as they are easily manipulated by hydration status, proximity to the last workout, and even digestive content.

Before/After Photos

Beyond showing quantitative progress, clients are often reinforced by actually visualizing their progress. Instructing clients to take photos of their body over time is simple and readily obtainable with only the need of a basic camera. However, this should be performed with consideration of consistency in conditions, angles, outfit, and lighting.

Conditions refers to taking body images at the same time and under the same conditions each time. For example, if a client takes baseline images first thing in the morning, after skipping dinner the evening before, but then takes the follow-up images on a full stomach or after drinking a lot of water, these follow up images can lead to a perceived lack of results when meaningful progress has been made. If a client wants to flex for baseline body images, they should be instructed to do the same for all follow-up photos.

Obtaining images at the wrong or inconsistent angle can misrepresent the actual appearance of the human body and limit the ability to effectively show progress. In general, clients should be recommended to use the following angles for body progress photos (**Figure: Body Progress Photos Angles**):

- Frontal: Directly facing the camera, standing with normal, relaxed posture
- Side: Facing perpendicular to the camera, standing with normal, relaxed posture
- Back: Facing away from the camera, with normal, relaxed posture



FigureBody Progress Photos Angles
Coach's Corner

Clients can also take a video with the held poses and then screenshot the image if they are having difficulty with timed photos.

Consistency in lighting is also important to obtaining a high-quality photo. In general, natural lighting is preferred but it is more important to maintain consistency. If a client feels more comfortable taking body images in the privacy of their bathroom, a fitness professional should recommend follow-up images in the same location and lighting conditions. Clients should be instructed to avoid any shadows across the body for all images.

Clients may share the body image results via e-mail or text message or upload into a secure shared folder set up by the fitness professional. Fitness professionals should organize client folders for secure photo storage, titled by date. To optimize the ability to show consistent progress, a regular schedule, such as taking body images on the first of each month, should be maintained. As significant progress is shown, putting together a quick collage of photos over time can be an invaluable tool for providing reinforcement and maintaining client motivation. Photos and measurements can also be more telling of body composition changes at a time when weight is stable or being influenced by outside forces (i.e., stress, fluid retention, etc.).

Coach's Corner

Getting consent from a client is important if the fitness professional wants to use client images for other reasons besides training.

Administering Movement and Mobility Assessments Virtually

In addition to quantitative assessments including range of motion, fitness professionals should also prompt the client to report any specific or broad areas of pain or discomfort experienced during movement and mobility assessments. Many clients may not have knowledge of physiological landmarks. Therefore, it may be best to have them point to the area of discomfort and rate their pain on a standardized scale (such as 0–10) and refer them to the appropriate practitioner (doctor or physical therapist) on a case-dependent basis, making sure to operate within the fitness professional's scope of certification. Certain applications may have a survey capability or self-administered assessment where clients can indicate the areas and associated levels of pain/soreness/discomfort. Such tools/functionality can be leveraged to monitor and track changes in body discomfort and soreness over time, which may be particularly useful to adjust and modify programming, if necessary, as well as provide reinforcement for clients who are looking to feel better as a program goal.

Apley Scratch Test

The Apley Scratch Test is a valuable assessment of upper extremity mobility limitations for consideration when developing a safe and effective exercise program. This assessment should only be performed under the live guidance of a qualified fitness professional using video conferencing. Fitness professionals should record this assessment for later interpretation to focus real-time efforts on guiding the client through a safe and effective assessment.

To perform a remote Apley Scratch Test (**Figure: Apley Scratch Test Technique**), consider the following:

- The client may be seated or standing to perform this assessment.
- The client should ask for help from a friend or set up their camera to ensure stability. The camera (phone, computer camera, or dedicated camera) should be level, and the client should be close enough to clearly visualize the area between the top of the head and lower back.
- Using the right arm, the client is instructed to touch their opposite shoulder blade with the arm behind the head and neck.
- The client is instructed to place the left arm behind the back and slowly reach upwards as far as possible with the goal of touching or even interlocking the fingers of both hands, if possible.
- The test is then repeated performing the opposite motions with each arm to assess and compare mobility in each arm/shoulder and between sides.



FigureApley Scratch Test Technique

Overhead Squat Assessment

The overhead squat assessment is a reliable functional evaluation of kinetic chain movement quality, neuromuscular control, and compensatory mechanisms that can be safely performed in a remote capacity with most client types. This assessment should only be performed under the live guidance of a qualified fitness professional using video conferencing or if the instructions are explicit and clear. It may be recorded by the client and sent to the fitness professional for observation and analysis. Fitness professionals should record this assessment for later interpretation and comparison to focus real-time efforts on guiding the client through the process safely and effectively.

While the overhead squat assessment may be used—and useful—for all client types, research has shown that it may provide more information in less-experienced clients and athletes who are still learning to control complex movement patterns (Bishop et al., 2016).

To perform an overhead squat assessment remotely (**Figure: Overhead Squat Technique**), consider the following:

- The client should ask for help from a friend or set up their camera to ensure stability. The camera should be level, and the client should be framed to capture the top of the fingertips down to a clear visual of both feet and ankles. Excess dead space above or below these physical landmarks should be avoided.
- Ideally, the client should wear form-fitting clothing within their level of comfort that does not match or blend into their background. A solid color and complete background (a wall) is preferable for observation and interpretation.
- Clients should be instructed to remove their socks and shoes for a clear visual of the feet and ankles.
- With the feet positioned shoulder-width apart and feet pointed directly forward, the client should raise both arms completely overhead with the elbows fully extended.
- Instruct the client to maintain arms in this overhead position, to the best of their ability, while squatting to a depth that brings their upper legs parallel to the ground (roughly to the height of a typical chair) or as deeply as possible without significant discomfort for a total of five repetitions.
- The assessment should be repeated with identical instructions from anterior, posterior, and lateral vantage points. To standardize the assessment, the camera should be left in the same place and the client instructed to simply rotate their body in relation to the camera to change angles.



FigureOverhead Squat Technique

NASM provides a comprehensive checkpoint table to assist fitness professionals in the standardized interpretation of movement compensations during the OHS, overactive, and underactive musculature and corrective program development.

Single-Leg Squat Assessment

The single-leg squat (SLS) is a simple, reliable, and relatively easy to administer assessment of stability, strength, balance, and motor control of the lower extremity, and Lumbo-Pelvic-Hip complex regions. Due to the complexity, strength, and balance requirements to perform an SLS assessment, it has been shown that physically active individuals with greater hip and knee flexion seem to be at lower risk to perform the SLS than non-physically active individuals (Gianola et al., 2017). The SLS assessment should be performed at the fitness professional's discretion based on the physical activity levels and injury history of the client. Similarly to the OHS, this assessment should be conducted during a live video conference and recorded for later interpretation in order to focus real-time efforts on guiding the client through the assessment safely and effectively. Conversely, if the instructions are explicit and clear, it may be recorded by the client and sent to the fitness professional for observation and analysis.

To perform an SLS assessment remotely, Livengood, Dimattia & Uhl (2004) recommends the following (**Figure: Single-leg Squat Technique**):

- The client should ask for help from a friend or set up their camera with live video conferencing to ensure stability. The camera should be level, and the client should be framed to capture the top of the head down to a clear visual of both feet and ankles. Excess dead space above or below these physical landmarks should be avoided.
- Ideally the client should wear form-fitting clothing, within their level of comfort, that provides visualization of both knees during all portions of the SLS assessment and that does not match or blend into their background. A solid color and complete background (a wall) is preferable for observation and interpretation.
- Clients should be instructed to remove their socks and shoes for a clear visual of the feet and ankles.
- Instruct the client to place their hands on their hips and to stand on a single leg with the other leg lifted off the ground, hip and knee flexed at a 45- and 90-degree angle, respectively. The foot of the non-stance leg should remain directly beside the stance leg/foot without touching it.
- While maintaining hands on hips, the client should squat to approximately 60 degrees of knee flexion or as deeply as possible without losing balance before returning to the start position in less than 6 seconds.
- Perform up to five repetitions before switching sides.
- Repeat identical assessment protocol on the other leg to assess for compensations and differences bilaterally.



FigureSingle-Leg Squat Technique

Fitness professionals should consider a modified single-leg balance assessment for clients who are not confident in their abilities, pregnant, obese, or exhibit any other contraindications to performing an SLS assessment.

NASM provides a comprehensive checkpoint table to assist fitness professionals in the standardized interpretation of movement compensations during the SLS, overactive, and underactive musculature and corrective program development.

Pushing/Pulling Assessments

Pushing and pulling assessments are a noninvasive method for assessing movement efficiency, mechanics, and muscle imbalances of the scapulae and shoulders and stability of the lumbo-pelvic-hip complex, cervical spine, and head. These assessments often require access to training equipment, such as free weights, a pulley system, or resistance bands, and may not be applicable or available for all remote-training clients. Ideally, the assessment should be performed live using video conferencing and recorded for later analysis and interpretation in order to focus real-time efforts on guiding the client through a safe and effective assessment. However, if the instructions are explicit and clear, and/or a video example with instructions is provided to the client, it may be a viable option for the client to record themselves performing the movements and send the recording to the fitness professional for observation and analysis. For the purposes of virtual assessments, variations utilizing a resistance exercise band are as follows.

To perform remote pushing and pulling assessments (**Figure: Pushing/Pulling Techniques**), consider the following:

- The client should ask for help from a friend or set up their camera (with live video conferencing) to ensure stability. The camera should be level, and the client should be framed to capture the top of the head down to a clear visual of both feet and ankles. Excess dead space above or below these physical landmarks should be avoided.
- It is important that the client wear form-fitting clothing within their level of comfort that does not match or blend into their background. A solid color and complete background (a wall) is preferable for interpretation.
- Choose a resistance band that allows for a minimum of 20 repetitions in each assessment. These assessments are not meant to assess muscular strength or cause significant fatigue.
- The client should be instructed to find a stationary object, such as a pole, to anchor the resistance band around ensuring equal length and tension on each side of the band. If the client has a door anchor specifically for use with resistance bands, they can certainly use that placed between the closed door and door jamb. Do not use an object that may move or fall over during this assessment.



FigurePushing/Pulling Techniques

- Holding a handle in each hand at the level of the mid-chest, the client should stand with feet shoulder-width apart, body centered to the stationary object, and toes pointing directly forward.
    - For the pushing assessment, the client should be facing away from the stationary object.
    - For the pulling assessment, the client should be facing toward the stationary object.
- While maintaining a controlled tempo throughout, the client should slowly push (or pull) handles away from (or toward) the body and return to the starting position. Approximately 10 to 20 repetitions should be completed for each pattern.
    - The pushing assessment is similar to a standing chest press.
    - The pulling assessment is similar to a standing row.
- Note any flaws in technique and movement compensations that should be addressed in the client's program.

Fitness professionals should use their judgement when deciding whether to perform pushing or pulling assessments with their virtual/remote clients.

Administering Muscular Endurance Assessments Virtually

Muscular endurance assessments are simple and generally safe due to the use of body weight alone and do not require additional equipment to perform. While it may still be preferable to perform muscular endurance assessments via video conference for accountability, instruction, and monitoring, these can generally be performed with basic instructional videos or e-mailed PDF guides and client reporting. Fitness professionals should consider running clients through a low-intensity, dynamic warm-up prior to conducting muscular endurance assessments.

Push-Up Tests

Push-up tests provide a strength and endurance assessment of the upper body musculature during a pushing pattern. While there are multiple variations available, the most common include maximal and modified push-up tests, which may be preferable for apparently healthy populations, including women, youth, obese, or older clients.



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To perform a virtual maximal push-up test, consider the following:

- Instruct the individual to assume a push-up position with hands slightly outside of shoulder-width apart, elbows and knees fully extended, and the spine in a neutral position. If the client is not able to perform a standard push-up, the client can perform push-ups from a kneeling position as outlined below.
- The client lowers their body until the elbows reach approximately 90 degrees of flexion or the chest makes contact with the ground.
- With no rest in between, return to the starting position with the arms fully extended.
- Continue until the client is no longer able to complete a repetition to the appropriate depth or fails to perform repetitions without compensation or any additional part of the body makes contact with the ground.
- Record the total number of repetitions completed.
- The client should be reassessed at regular intervals (e.g., 4 to 6 weeks) to evaluate progress.

To perform a virtual modified push-up test, consider the following:

- Instruct the individual to assume a push-up position with hands slightly outside of shoulder-width apart, elbows fully extended, knees on the ground hip-width apart, and the spine in a neutral position, with hips in line with the knees and shoulders.
- The client lowers their body until the elbows reach approximately 90 degrees of flexion or the chest makes contact with the ground.
- With no rest in between, return to the starting position with the arms fully extended.
- Continue until the client is no longer able to complete a repetition to the appropriate depth or fails to perform repetitions without compensation or any additional part of the body makes contact with the ground.
- Record the total number of repetitions completed.



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While clients may perform this on their own and self-report results or record their performance for the fitness professional to view later, ideally, this assessment should be performed live using video conferencing and recorded for later analysis and interpretation in order to focus real-time efforts on guiding the client through the assessment safely and effectively. The client should be reassessed at regular intervals (e.g., 4 to 6 weeks) to evaluate progress.

Plank Test

The plank test provides insight into a client's core and upper extremity strength, stability, and endurance. This timed assessment requires a stopwatch, which comes pre-installed on most connected devices and computers in the clock application.

To perform a virtual plank test, consider the following:

- The client should begin in a prone position on a level surface with the elbows and forearms placed on the ground. The elbows should be placed directly under the shoulders, level and in-line with mid-sternum to lower pectorals, with the toes pointing downward toward the ground.
- Instruct the client to begin the stopwatch and then lift the hips until their lower back and pelvis is parallel with the ground. Encourage the client to maintain a neutral head and neck alignment with the eyes focused down toward the floor between the hands.
- Instruct the client to hold this position for as long as possible but stop if pain or severe discomfort arises.
- The timer should be stopped when the client reaches failure, which is defined as inability to keep the hips off the ground, or form and technique are compromised in an attempt to maintain the required position—evidenced by compensations such as consistent arching or slouching of the lower back, excessive rounding or drooping of the upper back, and dropping or extending of the head and neck.



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While this assessment is valuable, it may be difficult to standardize the results due to the subjectivity of failure. For this reason, a fitness professional may consider performing the plank test during a recorded video conferencing session for post-analysis to determine the actual duration achieved.

Wall Sit Test

The wall sit test is a simple assessment of lower-body strength endurance, primarily of the quadriceps, hamstrings, and glute musculature. This timed assessment requires a stopwatch, which comes pre-installed on most connected devices and computers in the clock application.

To perform a virtual wall sit test, consider the following:

- Instruct the client to locate a flat, non-slippery surface next to a sturdy wall.
- With their back against the wall, have the client slowly slide down into a seated position, moving the feet away from the wall until a comfortable position is found with the knees and hips bent at 90 degrees. Ensure the knees are positioned directly above the feet and the thighs are parallel to the ground. Mark or make note of this position and have them stand back up.
- With the feet shoulder-width apart, instruct the client to move the feet out to this measured or marked distance—keeping the feet flat on the ground. Arms should be crossed on the chest or extended straight out to the front of the body, parallel to the ground.
- Upon achieving this position, start the timer.
- The client should hold this position for as long as possible but stop if pain or severe discomfort arises.
- The timer should be stopped when the client reaches failure, which is defined as an inability to maintain the 90-degree hip and knee angle.



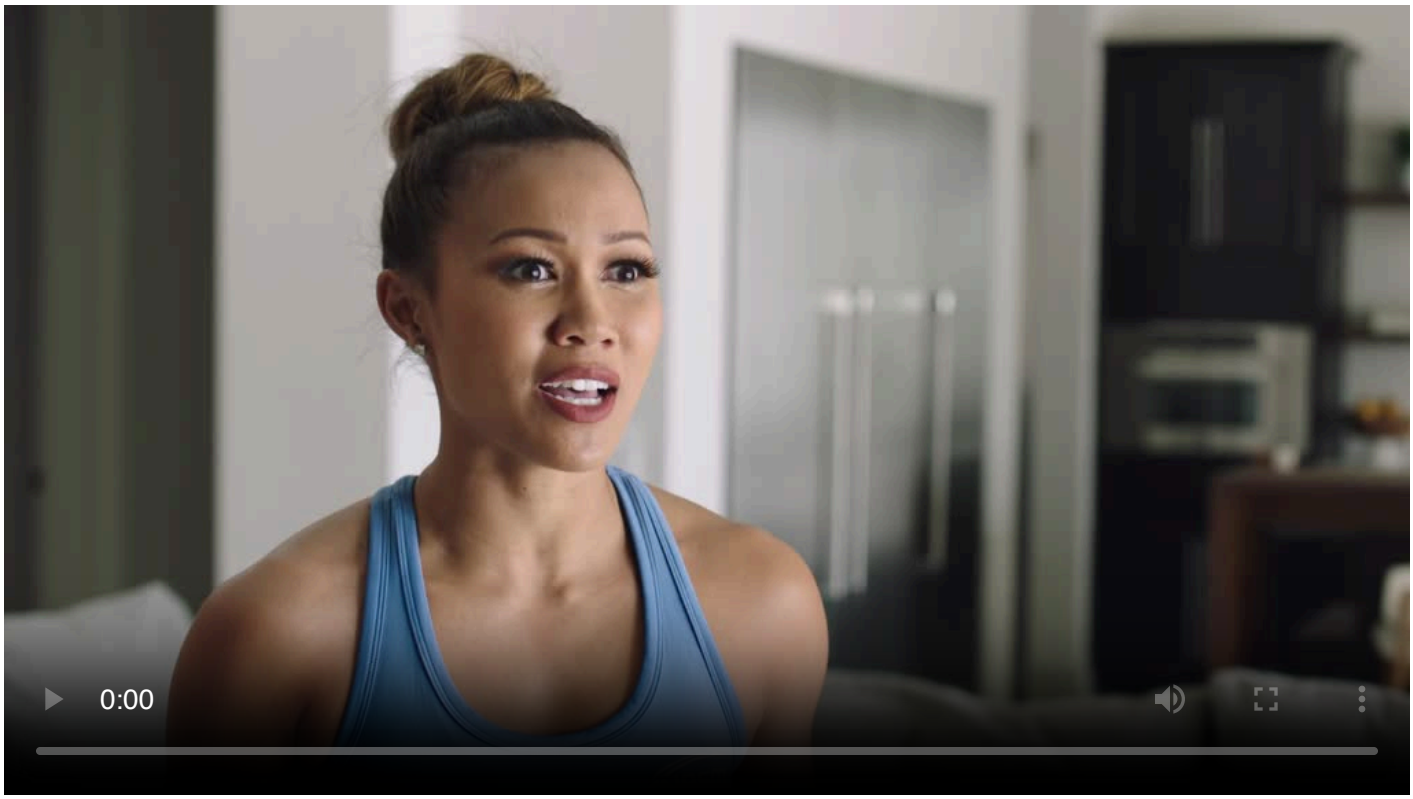
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Fitness professionals should use their judgement when determining whether to perform the wall sit test. It may not be suitable for certain individuals, particularly those with a history of knee pain or lower-body injury.

While clients may perform this on their own and self-report results or record their performance for the fitness professional to view later, ideally, this assessment should be performed live using video conferencing and recorded for later analysis and interpretation in order to focus real-time efforts on guiding and motivating the client through the assessment safely and effectively.

Summary

Administering virtual assessments may be a new concept to some fitness professionals. However, with proper planning; careful consideration toward assessment choices, standardized processes, and the use of readily available technology and resources; fitness professionals can build out their own comprehensive virtual assessment toolkit. This will streamline the assessment workflow while ensuring client comfort and safety, optimizing data collection and reliability, and, through regularly scheduled reassessments, allow for continued measurement and tracking of progress toward desired goals and outcomes. Please refer to the video below for a summary of the topics discussed within this chapter.



NASM VCS Lecture CH5
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